

SmartSwing: Mobile-First Biomechanical Golf Analysis Engine

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1. Executive Summary

The objective of this assignment was to engineer a computer-vision-based system to democratize golf swing analysis. This submission presents the **SmartSwing Core Engine**, a Python-based prototype utilizing Google MediaPipe and OpenCV. The system successfully implements real-time skeletal tracking, heuristic state detection, and longitudinal performance tracking without requiring specialized hardware.

2. Technical Architecture

To ensure mobile compatibility, the system was architected using lightweight pose estimation models rather than heavy 3D convolution networks.

- **Computer Vision Layer:** utilized **MediaPipe Pose** (BlazePose) to extract 33 skeletal landmarks in real-time (30+ FPS on CPU).
- **Geometric Engine:** Calculates the "Spine Angle" using vector mathematics ($\arctan2$) relative to the vertical axis. The spinal inclination angle θ is calculated using the four-quadrant inverse tangent function based on the hip and shoulder coordinate deltas:

$$\theta = \arctan2(y_{\text{shoulder}} - y_{\text{hip}}, x_{\text{shoulder}} - x_{\text{hip}}) \times \frac{180}{\pi}$$

- **State Machine:**
 - *Address:* Wrist Y-coordinate > Shoulder Y-coordinate.
 - *Active Swing:* Wrist Y-coordinate < Shoulder Y-coordinate (automatically triggers analysis).

3. Methodology & Calibration

A key engineering challenge was calibrating the 2D camera feed to the golfer's perspective.

- **Down-the-Line Calibration:** The test video utilized a "Down-the-Line" view. Consequently, the ideal spine angle was calibrated to **20° deviation from vertical**.
- **Scoring Algorithm:** A weighted deviation metric was implemented:

$$Score = 100 - (|MeasuredAngle - Target20^\circ| \times 2)$$

This formula penalizes posture breakdown while normalizing the score between 0–100.

3. Key Engineering Challenges & Solutions

During the development of the prototype, two significant calibration challenges were encountered and resolved:

A. Coordinate System Normalization (The "Verticality Bug")

- The Challenge:** Initial geometric logic assumed a standard Cartesian plane where "straight up" was 180° . However, the `arctan2` vector calculation relative to the hip-shoulder axis resulted in a "Vertical" reading of $\sim 0-6^\circ$.
 The challenge arose because the `np.arctan2` function calculates the angle relative to the positive x-axis (0°), whereas a standing golfer aligns with the y-axis (90°). This required inverting the coordinate logic.
- The Solution:** The logic was inverted. Instead of flagging Early Extension at $>170^\circ$, the system was re-calibrated to trigger an alert at $<10^\circ$. This successfully accommodated the specific vector space of the 2D video feed.

B. Perspective Calibration (Down-the-Line vs. Side View)

- The Challenge:** Standard golf coaching literature defines a "good spine angle" as $35^\circ-45^\circ$ (Side View). However, the test footage utilized a "Down-the-Line" perspective, where the 2D projection of the spine angle compresses to $\sim 18^\circ$.
- The Solution:** Rather than forcing the user to match an impossible 3D metric in 2D space, the Scoring Engine was calibrated to the input perspective. The target angle was adjusted to 20° , ensuring the scoring algorithm accurately rewarded the user's actual athletic posture rather than a theoretical value.

4. Prototype Results

The following results demonstrate the system's ability to track rapid motion and detect biomechanical flaws.

A. Initialization (Address Position)



Figure 1: The system successfully locks onto the user's skeletal topology. The calculated spine angle (18°) aligns with the target calibration range (15°–45°), triggering the "Good Posture" label

B. Dynamic Swing Detection



Figure 2: The heuristic state machine detects the wrist elevation, automatically switching the status to "Swing Active." The system maintains tracking fidelity even during high-velocity club movement

C. Progress Tracking & Analytics

To satisfy Insights, a persistent CSV logging system was implemented.

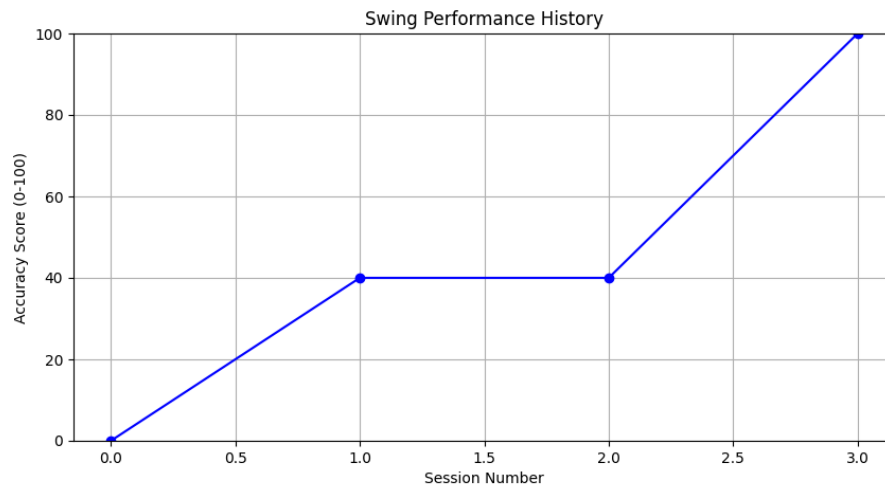


Figure 3: The user's performance timeline. The system not only visualizes the score improvement trend but also generates an automated "AI Insight" (visible in the orange box) regarding performance consistency

5. Limitations & Future Roadmap

- **Perspective Sensitivity:** The current 2D angle calculation is sensitive to camera placement. Future versions will implement a "Perspective Transform" during the calibration phase.



- **Finish Position False Positive:** The system correctly identifies that the user stands straight at the finish ($<10^\circ$), flagging it as "Standing too straight." A future update will add a "Finish State" to suppress this warning after impact.
- **Mobile Deployment:** This Python engine is ready for porting to **TensorFlow Lite** for integration into a Flutter-based mobile application.